EXERCISE 06

Poles and Zeros

RECAP

Where we left off...

- Last week we only needed one transformation pair. For an input $u(t) = e^{at}$:

$$\mathcal{L}\{e^{at}\} = \frac{1}{s-a} = U(s)$$

$\Rightarrow U(s)$ has a **pole** at $s = a$: it tells us which exponential the signal is made of.

$\Rightarrow$ For $s \neq a$, $U(s)$ is not 0: another exponential e^{st} is not totally different from e^{at}, it is still **similar** (both grow or decay, just at a different rate).

- Every SISO-LTI-system can be written as a fraction of two polynomials in frequency domain.

Time domain

$$\dot{x}(t) = Ax(t) + Bu(t)$$
$$y(t) = Cx(t) + Du(t)$$

Laplace-Transform

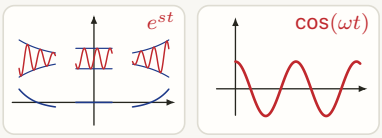
$$G(s) = C(s\mathbb{I} - A)^{-1}B + D$$

Frequency domain

$$G(s) = \frac{b_{n-1}s^{n-1} + b_{n-2}s^{n-2} + \cdots + b_0}{s^n + a_{n-1}s^{n-1} + \cdots + a_0} + d$$

- The variable s is a "complex frequency": $s = \sigma + j\omega$, where
  - $\mathrm{Re}(s) = \sigma$: **exponential** part
  - $\mathrm{Im}(s) = \omega$: **sinusoidal** part

$\Rightarrow$ In this course, most of the time we have $\sigma = 0$ (pure sinusoids).



1 | Sinusoids through a transfer function

Input signals as exponentials

- We saw, that most signals can be written as

$$u(t) = \sum_i U_i\, e^{s_i t} \qquad \Rightarrow \text{ EX. } \cos(\omega t) = \frac{1}{2}\left(e^{j\omega t} + e^{-j\omega t}\right)$$

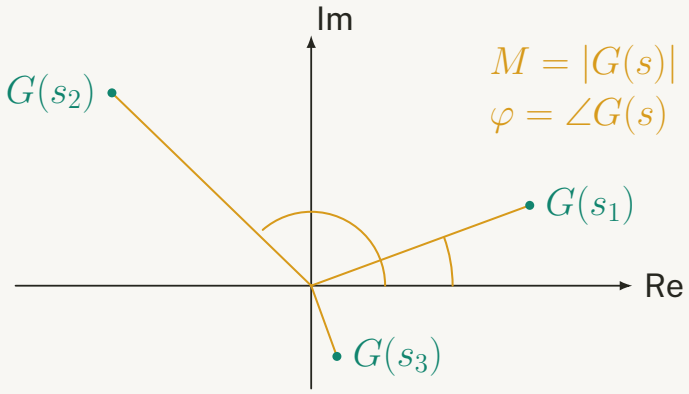
- Since we had that $u \longrightarrow \boxed{G(s)} \longrightarrow y$, we find that

$$y_{ss}(t) = \sum_i G(s_i)\, U_i\, e^{s_i t}$$

but if we evaluate this for some $s = \sigma + j\omega$, we get weird complex values...

$$\Rightarrow\ y_{ss}(t) = \frac{1}{2}\, G(j\omega)\, e^{j\omega t} + \frac{1}{2}\, G(-j\omega)\, e^{-j\omega t}$$

⇒ Plotting this for different s we can find:

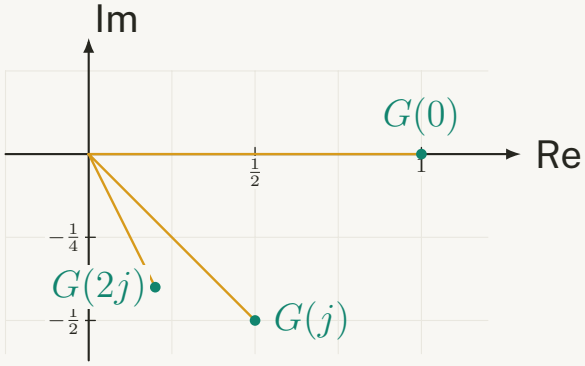


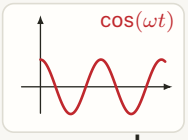
$M = |G(s)|$: Magnitude
$\varphi = \angle G(s)$: Phase

$$a + jb = M\, e^{j\varphi}$$

EX. Take $G(s) = \dfrac{1}{s+1}$ and plug in pure sinusoids $s = j\omega$:

s	$G(s)$	M	φ
0	1	1	$0°$
j	$\frac{1}{2} - \frac{1}{2}j$	$\frac{1}{\sqrt{2}} \approx 0.71$	$-45°$
$2j$	$\frac{1}{5} - \frac{2}{5}j$	$\frac{1}{\sqrt{5}} \approx 0.45$	$-63.4°$





$\Rightarrow G(s)$ evaluated at some s is just a complex number: a point in the complex plane with magnitude M and phase φ. Here: the higher the frequency ω, the **smaller** the amplitude and the **larger** the phase lag.

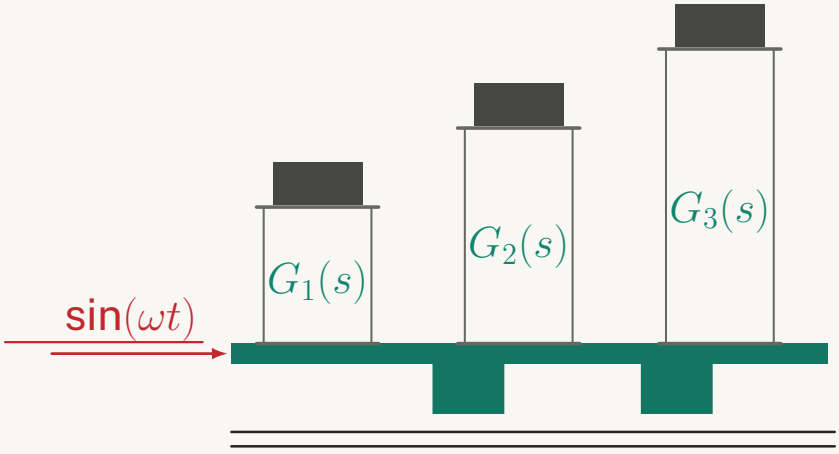
$\Rightarrow$ We can write

$$y_{ss}(t) = \tfrac{1}{2}\, \underline{M e^{j\varphi}}\, e^{j\omega t} + \tfrac{1}{2}\, \underline{M e^{-j\varphi}}\, e^{-j\omega t}$$

$$= M \cdot \tfrac{1}{2}\left(e^{j(\omega t+\varphi)} + e^{-j(\omega t+\varphi)}\right)$$

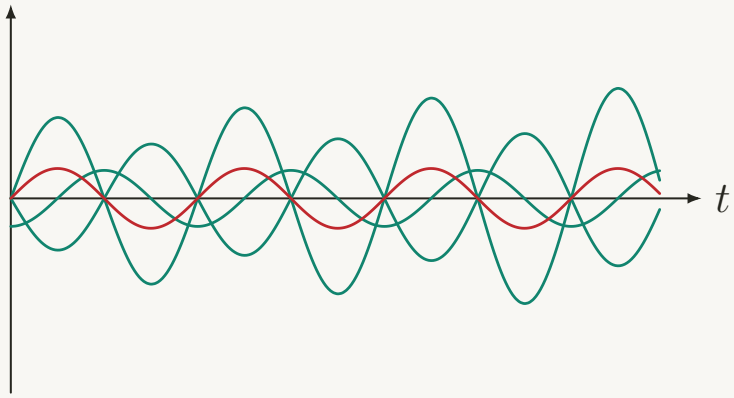
$\Rightarrow$ which is just $y_{ss}(t) = M\cos(\omega t + \varphi)$!!!

INTUITION

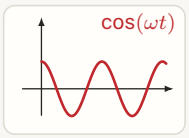


https://www.youtube.com/watch?v=LV_UuzEznHs

$\Rightarrow$ For the same s, these systems have different responses.



- Inputs experience a change in



- Amplitude: $|G(s_i)|$
- Phase: $\angle G(s_i)$,

depending on input frequency.

⇒ Output frequency does not change!!

Is that the whole story?

Now we have fully characterized systems in frequency domain, right? No...

- Take another look at the formula for the transfer function $G(s)$...

$$G(s) = \underline{C}(s\mathbb{I} - \underline{A})^{-1}\underline{B} + \underline{D}$$

⇒ Notice how we said a LTI-system is fully characterized by the following equation:

$$y(t) = \overbrace{\underline{C}e^{At}\,\underline{x(0)}}^{y_{ic}(t)} + \overbrace{\int_0^t \underline{C}e^{A(t-\tau)}\,\underline{B}\,u(\tau)\,d\tau \;+\; \underline{D}\,u(t)}^{y_F(t)}$$

⇒ All parameters included in TF formula, except: $x(0)$!!

⇒ TF's only show the **forced** response!

So far so good? Let's dive in-depth into TF-structure.

2 | Poles and zeros

Transfer functions: forms

- STANDARD FORM

$$G(s) = \frac{b_{n-1}s^{n-1} + b_{n-2}s^{n-2} + \cdots + b_0}{s^n + a_{n-1}s^{n-1} + \cdots + a_0} + d$$

- ROOT-LOCUS FORM

$$G(s) = \frac{k_H}{s^q} \cdot \frac{(s - z_1)(s - z_2)\cdots(s - z_m)}{(s - p_1)(s - p_2)\cdots(s - p_{n-q})}$$

- BODE FORM

$$G(s) = \frac{k_{Bode}}{s^q} \cdot \frac{\left(\frac{s}{-z_1} + 1\right)\left(\frac{s}{-z_2} + 1\right)\cdots\left(\frac{s}{-z_m} + 1\right)}{\left(\frac{s}{-p_1} + 1\right)\left(\frac{s}{-p_2} + 1\right)\cdots\left(\frac{s}{-p_{n-q}} + 1\right)}$$

- The scalar values p_i are called **poles** of $G(s)$
- The scalar values z_i are called **zeros** of $G(s)$

DEF Poles $\Rightarrow$ roots of the denominator of $G(s)$

Zeros $\Rightarrow$ roots of the numerator of $G(s)$

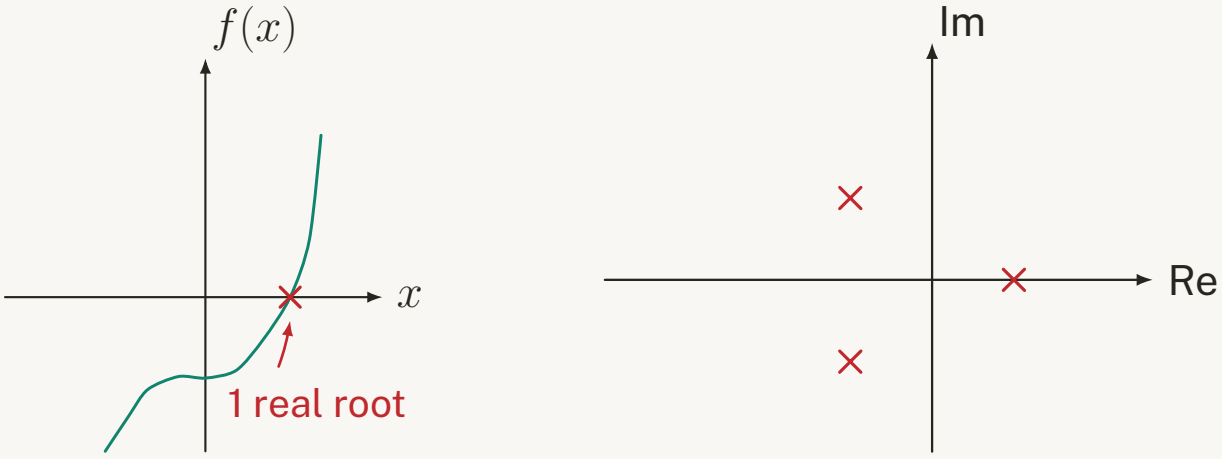
$$G(s) = \frac{N(s)}{D(s)} \quad \Rightarrow \quad \begin{aligned} z_i &: \ s_i \text{ s.t. } N(s) = 0 \\ p_i &: \ s_i \text{ s.t. } D(s) = 0 \end{aligned}$$

- Fundamental theorem of algebra:

  Order of polynomial $\widehat{=}$ # (complex) roots

Example: roots of a polynomial

EX. $f(x) = x^3 + x^2 - 2 = 0 \Rightarrow (x-1)(x+1+j)(x+1-j) = 0$



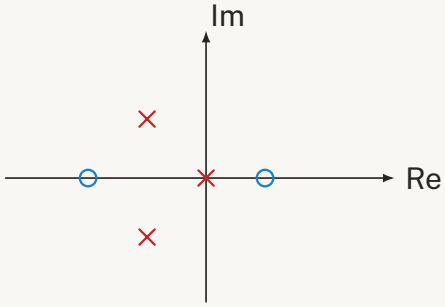
$\Rightarrow$ The location of poles & zeros are the defining characteristic of an LTI-system.

- Graphical notation (given root-locus form)

$$G(s) = \frac{k\,(s+2)(s-1)}{s\,(s+1+j)(s+1-j)}$$

$\times$: poles, $\circ$: zeros

$\rightarrow$



Poles

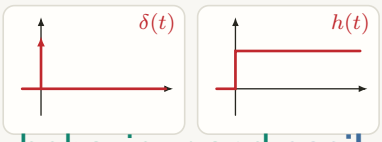
RECALL $G(s) = C(s\mathbb{I} - A)^{-1}B + D$

From linear algebra, we had that $A^{-1} = \dfrac{\text{adj}(A)}{\det(A)}$

$$\Rightarrow\ G(s) = \frac{1}{\det(s\mathbb{I} - A)} \cdot \overbrace{\left(C\,\text{adj}(s\mathbb{I} - A)B + D\det(s\mathbb{I} - A)\right)}^{N(s)} = \frac{N(s)}{\det(s\mathbb{I} - A)}$$

$\Rightarrow$ Setting $\det(s\mathbb{I} - A) = 0$ is how we find **eigenvalues**!

Poles of $G(s) \mathrel{\widehat{=}}$ Eigenvalues of A



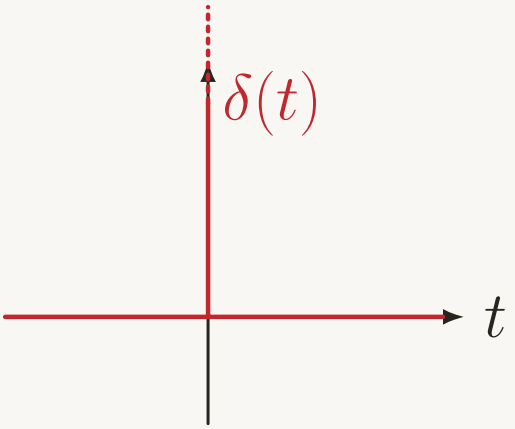
Position of poles will tell us about **stability**, **asymptotic** behaviour and **oscillatory** behavior!

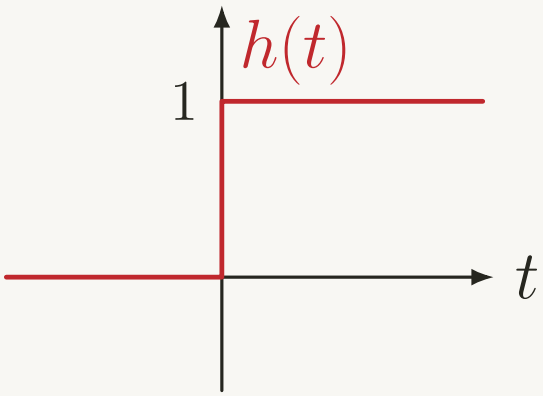
Test signals

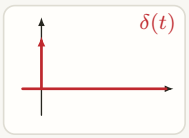
- To visualize the effect of the position of poles and zeros, we will use two fundamental input signals: $\Rightarrow$ **Test**-signals

TEST-SIGNALS (Some examples)

- **Dirac impulse**: $\delta(t) = \begin{cases} \infty, & t = 0 \\ 0, & \text{else} \end{cases}$
- **Heaviside** function: $h(t) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases}$ (**step** function)







3 | Effect of poles: the impulse response

$\Rightarrow$ Why are they relevant? Let's look at $\delta(t)$:

- The reaction of a system to a $\delta(t)$ input is called *Impulse-response*:

$$Y(s) = G(s)U(s) = G(s) \cdot \mathcal{L}\{\overset{1}{\delta(t)}\}$$
$$\Rightarrow\ Y(s) = G(s) \quad | \mathcal{L}^{-1}\{\cdot\}$$
$$\Rightarrow\ y(t) = \underbrace{\mathcal{L}^{-1}\{G(s)\}}$$

How do we find this?

$x(t)$	$X(s)$
impulse: $\delta(t)$	1
step: $h(t)$	$\frac{1}{s}$
$h(t) \cdot e^{at}$	$\frac{1}{s-a}$
$h(t) \cdot \sin(\omega t)$	$\frac{\omega}{s^2+\omega^2}$
$h(t) \cdot \cos(\omega t)$	$\frac{s}{s^2+\omega^2}$

- Partial fraction expansion

$$G(s) = \frac{b_{n-1}s^{n-1} + b_{n-2}s^{n-2} + \cdots + b_0}{s^n + a_{n-1}s^{n-1} + \cdots + a_0}$$

$$G(s) = \underbrace{\frac{r_1}{s-p_1}} + \underbrace{\frac{r_2}{s-p_2}} + \cdots + \underbrace{\frac{r_n}{s-p_n}}$$

r_i: residues

Of these terms we can easily find $\mathcal{L}^{-1}\{\cdot\}$

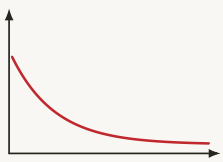
$$\Rightarrow\ y(t) = \mathcal{L}^{-1}\left\{\frac{r_1}{s-p_1} + \frac{r_2}{s-p_2} + \cdots + \frac{r_n}{s-p_n}\right\} \quad | \text{ Laplace is linear !}$$

$$\Rightarrow\ y(t) = r_1 e^{p_1 t} + r_2 e^{p_2 t} + \cdots + r_n e^{p_n t}$$

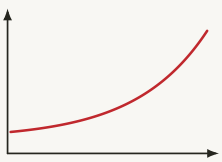
Now we can think of the effect of different poles.

RECALL $s = \sigma + j\omega$

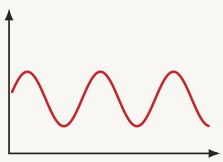
- $e^{\sigma t},\ \sigma < 0$



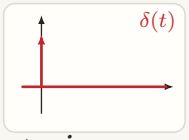
- $e^{\sigma t},\ \sigma > 0$



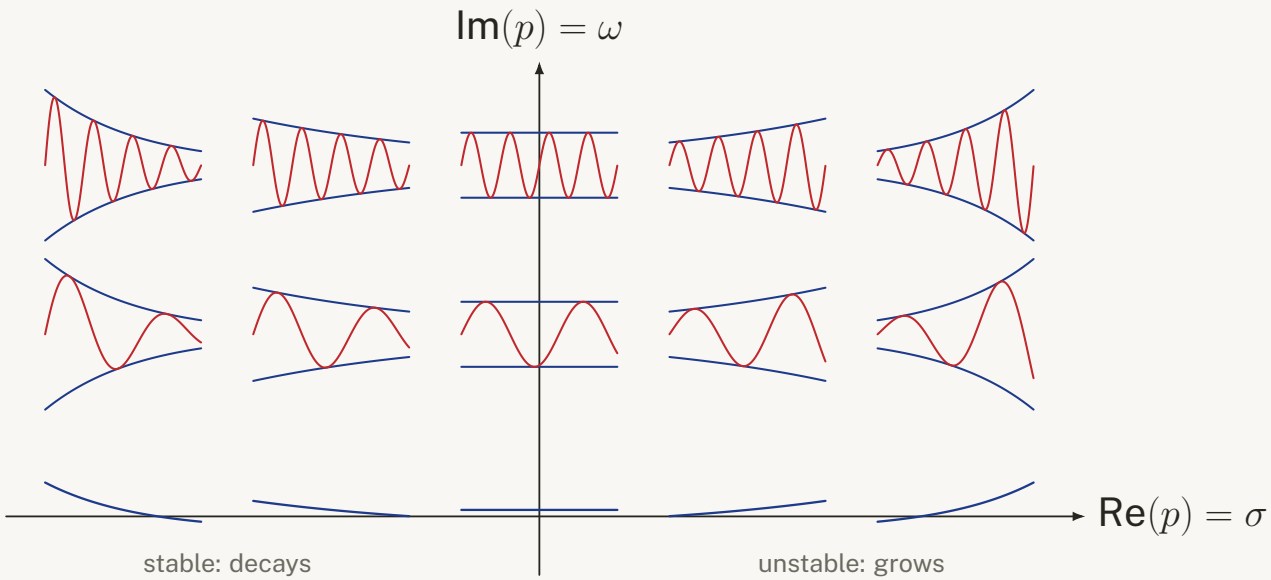
- $e^{j\omega t} \pm e^{-j\omega t}$



$\Rightarrow$ Matlab

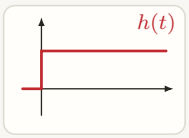


OVERVIEW Shape of the mode e^{pt} for different pole positions $p = \sigma + j\omega$:



$$Y(s) = \underbrace{G(s)}_{\text{system}} \cdot \underbrace{U(s)}_{\text{input}}$$

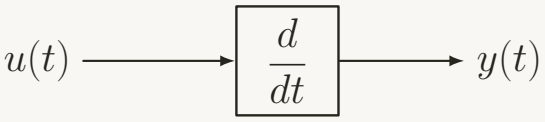
$\Rightarrow$ The poles p in this figure are the poles of the system $G(s)$: they fix its modes e^{pt}, no matter which input we apply.



4 | Effect of zeros

Zeros and derivatives

To understand the effect of zeros, consider this:



$$y(t) = \frac{d}{dt}u(t)$$
$$\Downarrow$$
$$Y(s) = \underbrace{s}\cdot U(s)$$

RECALL THIS PROPERTY:

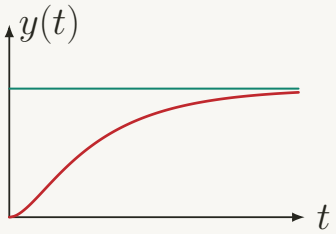
$$\mathcal{L}\{f^{(n)}(t)\} = s^n F(s)$$

New zero in TF!

$\Rightarrow$ Zeros have some connection to derivatives...

How does adding zeros to a system affect their responses?

- Let's try and see this with step responses
- Assume some system $G(s)$ without zeros and as input, apply a step $h(t)$:



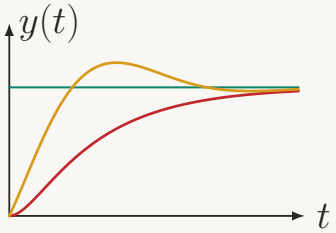
$\Rightarrow$ How would this change if we added a zero?

$$\tilde{G}(s) = \left(\frac{s}{-z} + 1\right)G(s)$$
$$= \frac{s}{-z}G(s) + G(s)$$

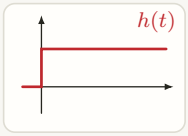
$\Rightarrow$ The "s" acts as a derivative so the output would look like:

$$\tilde{y}(t) = -\tfrac{1}{z}\dot{y}(t) + y(t)$$

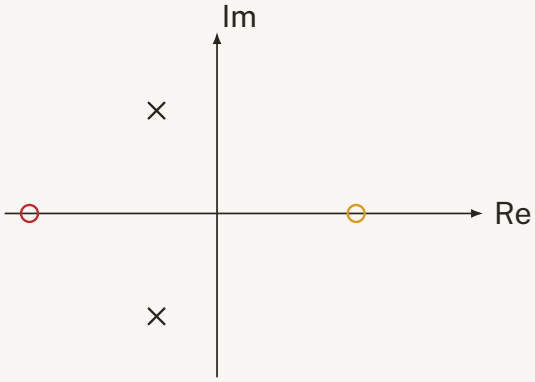
We add on top a **scaled** derivative of $y(t)$!

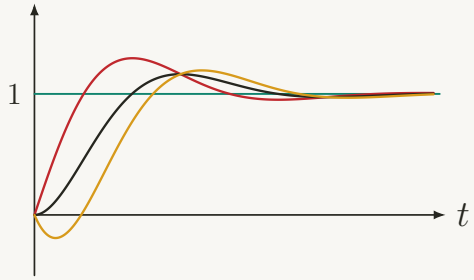


The zero has **anticipatory** effect! It "knows in advance" that it is off from the goal.

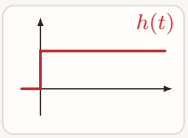


- Placement of zeros





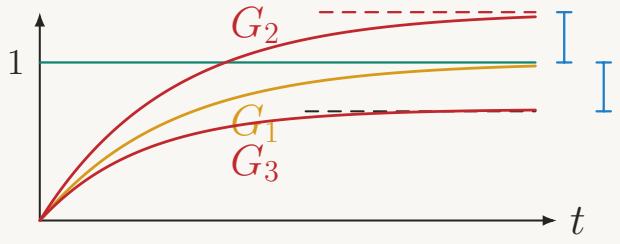
- Zeros in left-half plane: **Minimumphase** zeros
- Zeros in right-half plane: **Non**-minimumphase zeros
  - Think of how the location of z affects $\tilde{y}(t) = -\frac{1}{z}\dot{y}(t) + y(t)$



5 | Steady state: the final value theorem

- The step response is a really useful tool to analyze systems.

⇒ One thing that we have glossed over...



- A system doesn't necessarily converge to the input value!

$$y_{ss}(t) = \lim_{t\to\infty} y(t) \neq u(t)$$

- So how could we quantify this **offset**?

Final Value Theorem

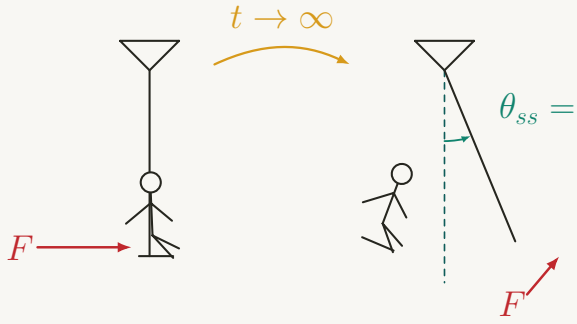
$$\lim_{t\to\infty} y(t) = \lim_{s\to 0} sY(s) = \lim_{s\to 0} sG(s)U(s)$$

$x(t)$	$X(s)$
impulse: $\delta(t)$	1
step: $h(t)$	$\frac{1}{s}$
$h(t) \cdot e^{at}$	$\frac{1}{s-a}$
$h(t) \cdot \sin(\omega t)$	$\frac{\omega}{s^2+\omega^2}$
$h(t) \cdot \cos(\omega t)$	$\frac{s}{s^2+\omega^2}$

- Step function: $y_{ss} = \lim_{s\to 0} sG(s) \cdot \overbrace{H(s)}^{1/s}$

$$y_{ss} = \lim_{s\to 0} \cancel{s} G(s) \cdot \cancel{1/s} = \lim_{s\to 0} G(s)$$

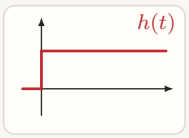
Example: step force on a swing



$$\dot{x}(t) = \begin{bmatrix} 0 & 1 \\ \frac{g}{\ell} & -\frac{c}{m\ell^2} \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ \frac{1}{m\ell} \end{bmatrix} u(t)$$

$$y(t) = \begin{bmatrix} 1 & 0 \end{bmatrix} x(t)$$

$$F(t) = \begin{cases} 0, & t < 0 \\ 1, & \text{else} \end{cases}$$



- We could now solve this as

$$\theta_{ss} = \lim_{t\to\infty}\left(Ce^{At}x_0 + \int_0^t Ce^{A(t-\tau)}B\,d\tau\right)$$

Good luck...

Better idea: transform to frequency domain!

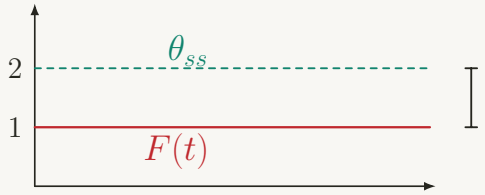
$$G(s) = C(s\mathbb{I} - A)^{-1}B + D = \frac{4}{s^2+s+2}$$ | Random params

Then use final value theorem

$$\theta_{ss} = \lim_{s\to 0} \cancel{s}\,G(s)\,\cancel{U(s)}, \qquad U(s) = H(s) = \tfrac{1}{s} \text{ (step-function)}$$

$$= \lim_{s\to 0}\frac{4}{s^2+s+2} = \frac{4}{2} = 2 \;//$$

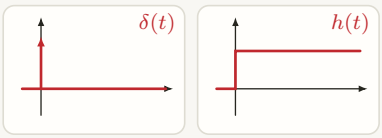
(Degrees, radians,...)



This factor has a name...

$\Rightarrow\ G(s\to 0) = k_{Bode}$

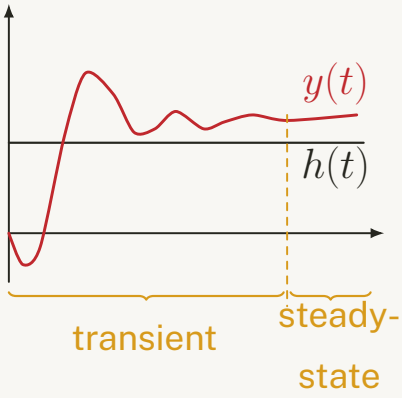
"Bode gain"



6 | Summary and a warning

In summary

$\Rightarrow$ If you have a system $G(s)$ and want to know how it responds to a step/impulse



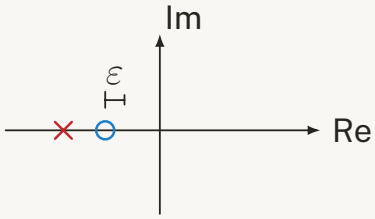
Transient: **Poles/zeros**

Steady-state: k_{Bode}

Pole-zero cancellation

- Consider a system with a zero near a pole

$$G(s) = \frac{s+1+\varepsilon}{(s+1)(s+2)}$$



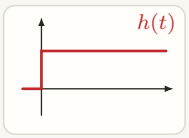
what happens if $\varepsilon \to 0$?

$$G(s) = \frac{\cancel{s+1}}{\cancel{(s+1)}(s+2)}$$

The pole and the zero seem to cancel!

But they dont!!!

The internal dynamics remain, we just don't see them anymore in the output! Dangerous if **unstable**.



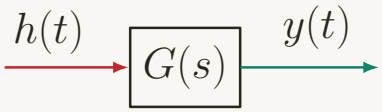
Integrators and differentiators

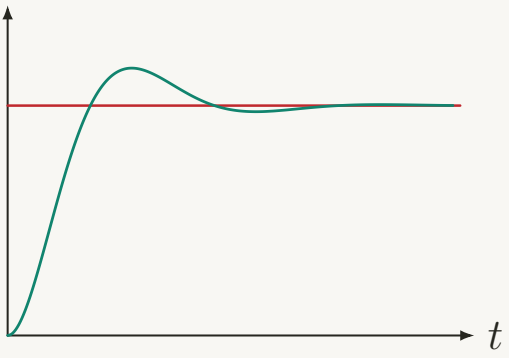
Poles at $p = 0$: Integrators

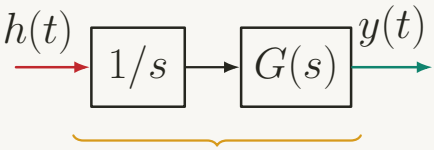
Zeros at $z = 0$: Differentiators

$\Rightarrow$ We will use this in controller design

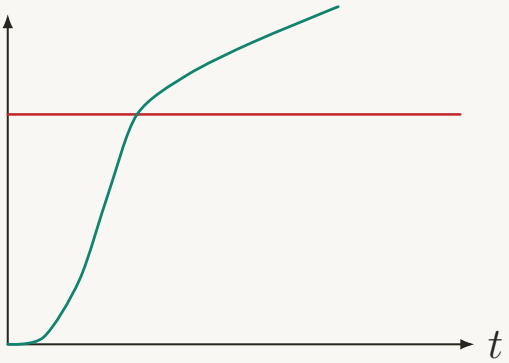
$$G(s) = \frac{1}{s^2 + s + 1}$$

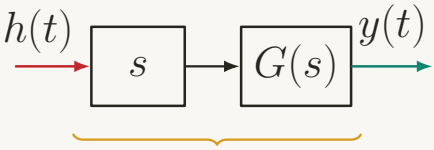




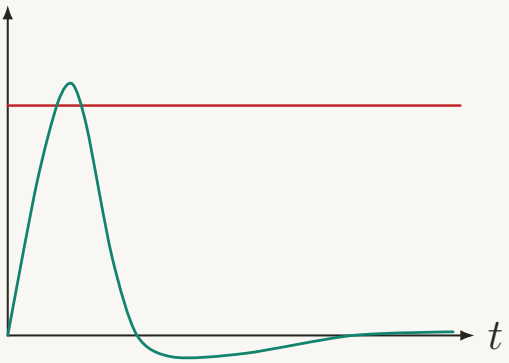


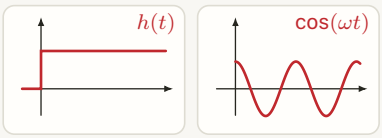
$$\frac{1}{s^3 + s^2 + s}$$





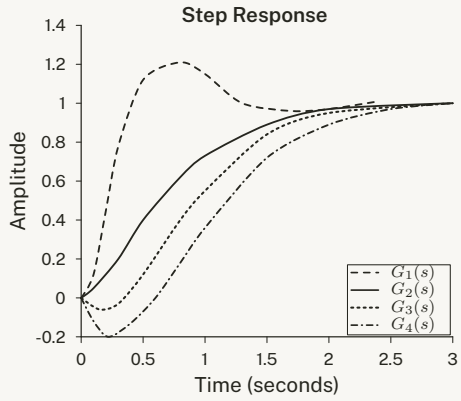
$$\frac{s}{s^2 + s + 1}$$

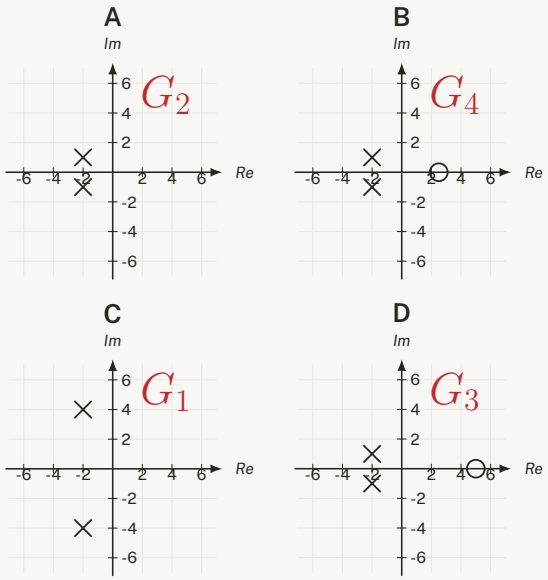




QUIZ

Q1) Match poles/zeros to step response





$\tilde{y} = y + \frac{1}{-z}\dot{y}$

Q2)

Problem: Consider a linear time-invariant system with transfer function $G(s)$ that is given by a root-locus gain $k_{rl} = 1$ and by two poles and two zeros as shown in Figure 7.

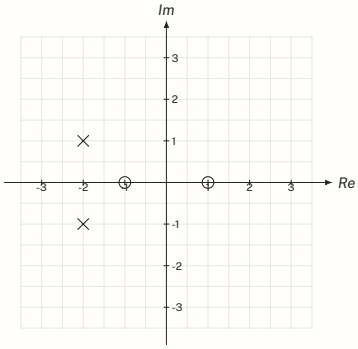


Figure 7: Poles ($\times$) and zeros ($\circ$) of the LTI system $G(s)$.

Q19 (1 Points) Mark the correct answer.

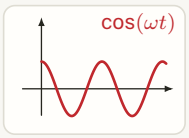
What is the magnitude $|G(j)|$ of $G(s = j)$?

[X] $|G(j)| = \frac{1}{2\sqrt{2}}$ [B] $|G(j)| = \frac{1}{4}$ [C] $|G(j)| = \frac{1}{2}$ [D] $|G(j)| = \frac{1}{\sqrt{2}}$

Q20 (1 Points) Mark the correct answer.

What is the phase $\angle G(j)$ of $G(s = j)$?

[A] $\angle G(j) = 45°$ [B] $\angle G(j) = 0°$ [C] $\angle G(j) = -45°$ [X] $\angle G(j) = 135°$



SOLUTION

Poles: $-2+j,\ -2-j$

Zeros: $1,\ -1$

$$G(s)=\frac{(s-1)(s+1)}{(s+2-j)(s+2+j)}$$

$$G(s=j)=\frac{(j-1)(j+1)}{2(2+2j)}=\frac{-2}{4+4j}=-\frac{1}{2+2j}$$

- $$|G(s=j)|=\frac{|-1|}{|2+2j|}=\frac{1}{\sqrt{(2)^2+(2)^2}}=\frac{1}{\sqrt{8}}=\frac{1}{2\sqrt{2}}$$ //

- $$\angle G(s=j)=\angle\frac{-1}{2+2j}=\angle -1+\angle\frac{1}{2+2j}$$

$$=\pi+\cancel{\angle 1}-\angle(2+2j)=\pi-\tan^{-1}\left(\tfrac{2}{2}\right)$$

$$=\pi-\tfrac{\pi}{4}=\tfrac{3\pi}{4}=135^\circ$$ //